Western Regional Collegiate Cyber Defense Competition

Qualifiers Team Packet

February 8, 2025

Welcome to Steve's Crazy Pizza! **3**
Getting Started **3**
Our Culture **4**
What You'll Be Doing **4**
The Cool Things About Us and What We Do **5**
Restaurant Stuff 5
Definitely Not Shady Organization Stuff 5
Pizza As A Service 5
Continuous Pizza Delivery and Ingestion (CD/CI) 5
Games Room 5
Environment and Network Topology **6**
Topology Diagram 6
Services 7
Router 7
Recruitment 7
Domain Manager 7
Source Code Management System 7
PR Sites 7
Toppings-Suggesting Bot 7
“Toy” Shop 8
Blogs and Websites 8
Pizza Ordering System 8
CCTV 8
CD/CI 8
Western Regional CCDC Mission and Objectives **10**
Competition Overview **10**
Competition Goals **10**
Connection Guidance **11**
Remote Testing 11
Limited vCenter Access 12
Support and Tickets **14**
Discord 14
Ticket Service 15
Your password: (See Password Document) 15
Submitting Tickets (Blue Team) 15
Common Service Requests 16
Inject Scoring **17**
This will be the only notice for inject scoring guidelines. Points will be deducted for not following these rules. 17
File Names: 17
Citing Sources: 18

Manual Scoring (Orange Team) 18
Orange Team Services (Initial) 18
Point Deductions (Red Team) 19
Incident Reports (Red Team) 20
Remote Site Judges 20
Proxy Server 20

Welcome to Steve's Crazy Pizza!

Getting Started

Hello [redacted] (IT administrators)!

Operating regionally, the [redacted] (Company) acts beyond conventional tastes, with the task of providing pizza for [redacted] (people).

We maintain an extensive database of information regarding our pizzas requiring Special Catering Procedures, and all our pizzas at Steve's Crazy PIzza always match the natural tastes that the people implicitly trust in.

We operate to maintain our pizza delivery, so that our customers can live and go on with their daily lives without fear, mistrust, or doubt in the Company's pizzas, and to maintain human [redacted] (meal satisfaction).

Our mission is threefold:

- Serve — serve unique pizzas, but keep their recipes confidential to prevent them from falling into the hands of civilian or rival [redacted] (Companies) through extensive observation and surveillance, acting to intercept [redacted] (leaks and adversaries) at the earliest opportunity.
- Cater — cater our pizza to spread our influence and effects; by [redacted] (serving them everywhere), or by [redacted] (supporting) public dissemination of knowledge thereof.
- Pizza — protects humanity as well as the [redacted] (customers) until such time that the pizzas are either fully understood or new theories of science can be devised based on their properties and behavior.

Additional information has been provided now that you have joined us in pursuit of our goals. Welcome to the [redacted] (Company), and good luck.

Steve's Crazy Pizza serves delicious pizza. *Where is Steve?*

Our Culture

We at Steve's Crazy Pizza entry positions proudly boast a mere [redacted] fatality rate, so expect an incredibly safe work environment! We love to guide you every step of the way, so keep in mind that you are always being watched by our encouraging supervisors. Here, we value our employee happiness so much that if you look at the wrong things you may be subject to [redacted] (vacation), so just remember if there seems like there's more than meets the eye there's nothing more than meets the eye!

What You'll Be Doing

Congratulations on being [redacted] (our new IT administrators)! In your position, you will be instructed to maintain Steve's Crazy Pizza's infrastructure. This serves as a potential major income stream for the [redacted] (Company), and hence you should:

- Comply with requests from the [redacted] (Company) relating to the maintenance of our infrastructure
- Defend against any malevolent [redacted] (customers) targeting our critical infrastructure
- Allow for and manage deployments as apart of our Continuous pizza Delivery and Ingestion
- Administering any services on the network, and handling business operations
- Providing support and assistance to any potential customers

If any clearance-related discussion comes up, for all means and purposes your clearance level is currently Level 0, and your designation is "Class D Security Officer (Recruit)", however you may be given access to higher-level classified material at the discretion of [redacted] ("Steve"). Lying about this may result in immediate [redacted] (forced retirement).

The Cool Things About Us and What We Do

Restaurant Stuff

If you ever get hungry on the job, pizzas are on the house! Feel free to feast on our variety of delectable pizza!

Definitely Not Shady Organization Stuff

Working here, you will see more than you would see in any other job! Though, keep in mind, curiosity and inquisitive employees are subject to the aforementioned free vacation!

Pizza As A Service

We proudly provide employees with access to our painless pizza-ordering, "toys", and similar items, all available on their corresponding services on-demand. Feel free to get yourself a teddy bear if this job ever brings you to tears!

Continuous Pizza Delivery and Ingestion (CD/CI)

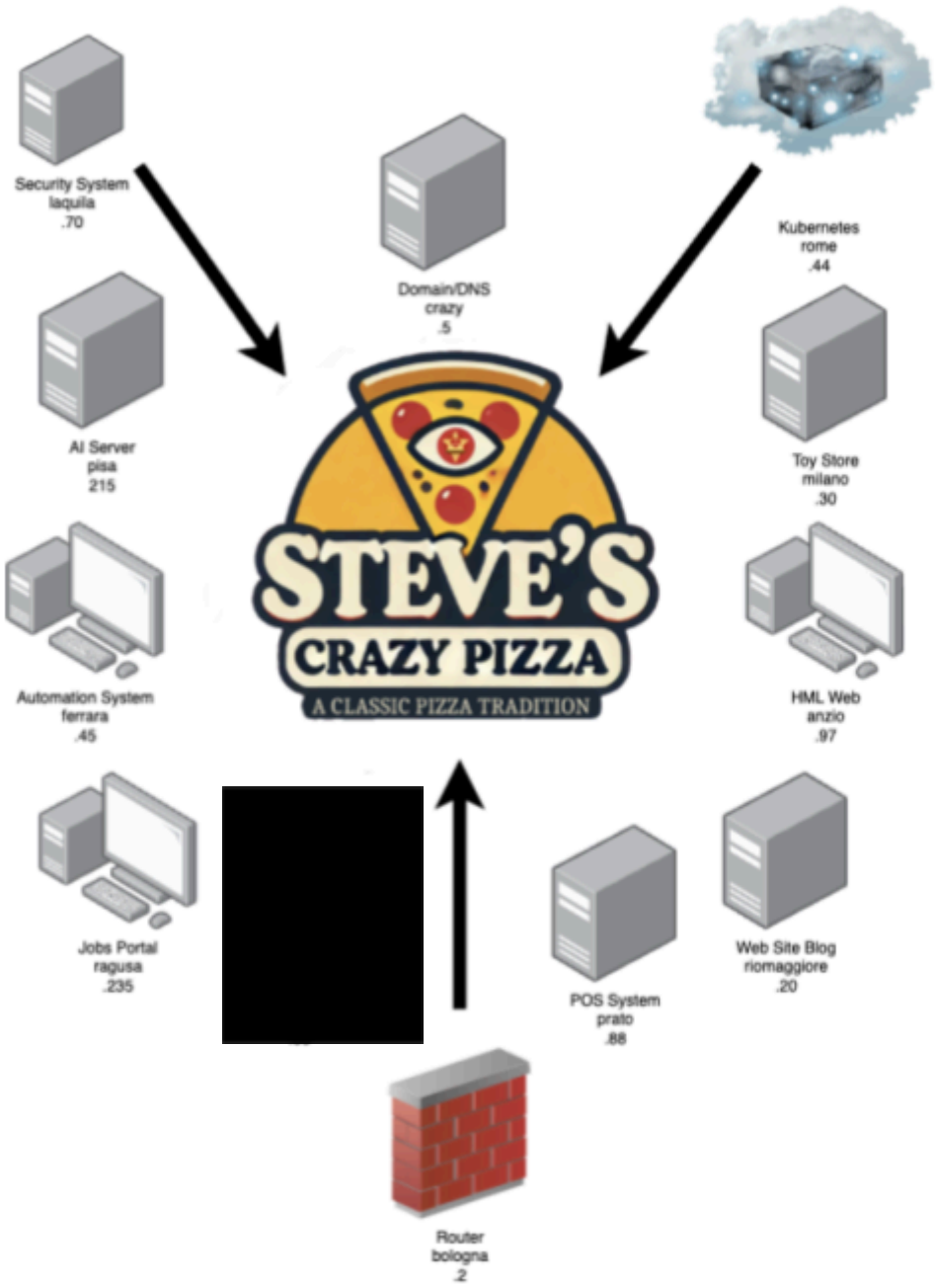
Some of our services have "CD/CI" systems setup, which allows for them to automatically update on sauce code changes. This is a critical part of our baking, and should remain functional, as it allows us to alternate between our [redacted] (employee & customer) menus, and our FDA approved menu (why is this empty…)!

Games Room

Imagine an endless maze of bright, buzzing fluorescent lights overhead, with soft, squishy carpet underfoot and the gentle hum of air conditioning creating a soothing ambiance. It's a playground where you can get lost in [redacted] of quirky, oddly satisfying rooms, each one inviting you to explore deeper into the infinite expanse. From open spaces to cozy nooks filled with strange, delightful puzzles, there's always a surprise waiting around every corner. We at Steve's Crazy Pizza want to make sure you have a chance to enjoy!

Environment and Network Topology

Topology Diagram



Services

Router

To quote our team, "it's a router", a router capable of securely handling all the incredibly-high bandwidth requirements of running our pizza company. Like all pizza company routers, you could [redacted] (handle complicated pizza orders) through this router!

Recruitment

Only the best minds have the mental fortitude to deal with the pepperoni and whatever else goes on pizza. It's imperative that we maintain full access to our recruiting sites, which look for the finest minds in areas normal to pizza making like cryptanalysis, [redacted] (foreign) languages, and more!

Domain Manager

Like most pizza companies, we have hundreds of employees enrolled, and are aiming to incorporate perhaps tens of thousands at a later date. Working with that kind of user management gets tedious, so we have a website for handling our domain!

Source Code Management System

Given our reliance on custom sites throughout our infrastructure [redacted] (because what kinda pizza companies trusts those big open-source projects that do the same thing), we have a publicly-facing source code management system for maintaining our projects, and [redacted] (so other innocent companies can learn from them)! We *require* all websites on our source code management system to have up-to-date source code that matches what is in production, especially for systems reliant on our CD/CI.

PR Sites

Because we [redacted] (love updating our community about our pizzas), and because we may occasionally need to release health warnings about our [redacted] (products), we have a cutting edge PR site!

Toppings-Suggesting Bot

We have found customers are frequently unaware of our hundreds of pizza toppings, and given this list is always changing we have implemented a toppings-suggesting bot, an LLM-guided bot to recommend our finest of toppings! We know you want those [redacted] (pickled pistachio shells)!!

“Toy” Shop

We have a site for [redacted] and merchandise! Who doesn’t love merchandise???

Blogs and Websites

Ranging from recipe sites, to legendary food reviews we have sites to share our love of pizza and [redacted]

Pizza Ordering System

We love pizza, and we sell pizza! We couldn’t *contain* our excitement for pizza, so to *protect* our customers from unintuitive pizza checkouts, we have implemented a *secure* one-click pizza ordering system!

CCTV

We have a CCTV system setup and website to handle our beautiful streams of [redacted] (the many doors and rooms of our sprawling pizza center)! We love watching [redacted] (our doors) for that extra security [redacted] (from hungry pizza customers)!

CD/CI

Our CD/CI system handles updating our sites rapidly so we can pass all of those nasty FDA inspections! We use it to periodically update our menus, our merchandise, you name it!

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Western Regional CCDC Mission and Objectives

The Western Regional Collegiate Cyber Defense Competition (CCDC) provides an opportunity for educational institutions to compete, and is part of a national organization (see www.nationalccdc.org) to provide a unified approach across nine regions of the country. Qualified educational institutions include those with information assurance or computer security curricula. The Western Regional Collegiate Cyber Defense Competition is designed to provide a controlled competitive environment that will permit each participating institution to assess their students' depth of understanding and operational competency in managing the challenges inherent in protecting an enterprise network infrastructure and business information systems.

Competition Overview

The Western Regional Collegiate Cyber Defense Competition (WRCCDC) is designed to test each student team's ability to secure a networked computer system while maintaining standard business functionality. The scenario involves team members simulating a group of employees from an IT service company that will initiate administration of an IT infrastructure. The teams are expected to manage the computer network, keep it operational, and prevent unauthorized access. Each team will be expected to maintain and provide public services: a website, a secure web site, an email server, a database server, an online curriculum server, and workstations used by simulated sales, marketing, and research staff as per company policy and mission. Each team will start the competition with a set of identically configured systems.

The objective of the competition is to measure a team's ability to maintain secure computer network operations in a simulated business environment. This is not just a technical competition, but also one built upon the foundation of business operations, policy, and procedures. Student teams will be scored on the basis of their ability to detect and respond to outside threats, including cyber attack while maintaining availability of existing network services such as mail servers and web servers, respond to business requests such as the addition or removal of additional services, and balance security against varying business needs.

Competition Goals

1. To evaluate the defensive and responsive skills of each team under exact hardware, software application, and operating system configurations using a joint academic and industry rating scale
2. To demonstrate the effectiveness of each participating institution's academic security program
3. To have industry recognition, participation and acceptance of each competition
4. To provide a cooperative and competitive atmosphere among industry partners and academia in the area of cyber defense education
5. To provide recognition for participating teams
6. To increase public awareness of academic and industry efforts in the area of cyber defense education

Connection Guidance

Teams will receive PAN Global Protect credentials for VPN access. Each team member will access the VPN with only one end device. Your device count is capped at 9 devices. This connection will serve as your starting point for ALL of your players. You basically will share the network (Via separate desktops / terminals.) Don't worry, it is powerful enough to support your activities for the day of the competition. You no longer need to supply your IPv4 addresses. We will monitor the number of addresses and connections into the environment. PAN Global Protect collects system fingerprint information to help us identify each student (or connect).

To help try to clarify, each competitor will have an account. ONLY one account can be used at a time. Each Team gets 9 accounts (8 Team Members and 1 Coach). Once in this network, they can reach their IP addresses relating to their pod via the connection. This will happen promptly at 9AM and you will have access onto the 192.168.220.0./ 24 network. This IP address range accepts credentials via SSH, VNC, Telnet, and Remote Desktop depending on device (See Table Above).

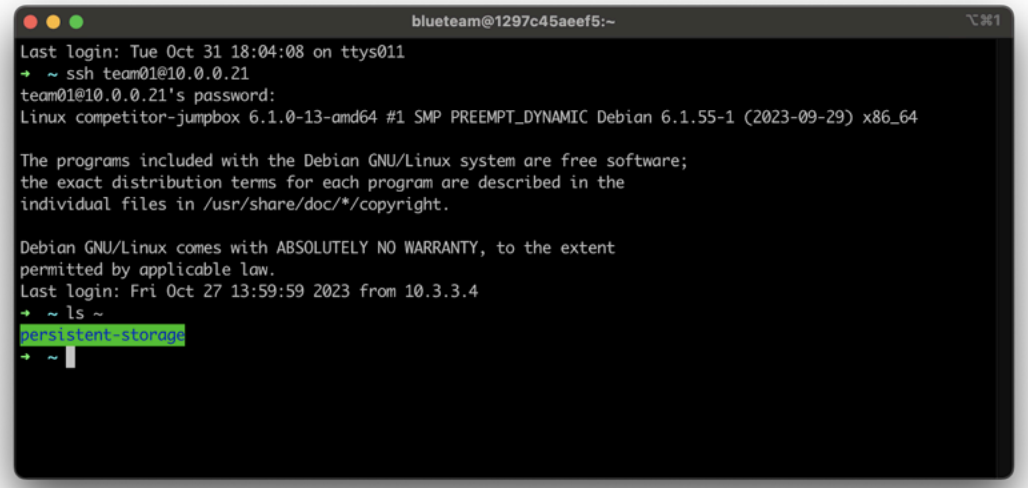
Remote Testing

To better provide teams with the ability to analyze and troubleshoot their environments a remote shell has been provided to teams. This allows you to test any common issues from connectivity, dns, service validation, and much more from a dedicated shell running outside of the competition environment. This system WILL NOT be attacked by the red team. There is a persistent storage volume so competitors can share files between other members of their team, any files not in that directory will be removed once they log out.

To access you can use your preferred *ssh client* to connect to jump.wrccdc.org or 10.0.0.21. To log in use the following:

Your competition username: teamXX (where team XX is 01…04..10, 25, etc)

Your password: (See Password Document)



A complete list of tools is available at our GitHub https://github.com/wrccdc-org/competitor-container. Use this system to perform any tests you need against systems in your environment. Please make sure to use the public IP Addresses, such as 10.100.1YY.ZZ. Where YY is your team number, and ZZ is the last octet of the system. If you have questions please reach out to the Operations Team.

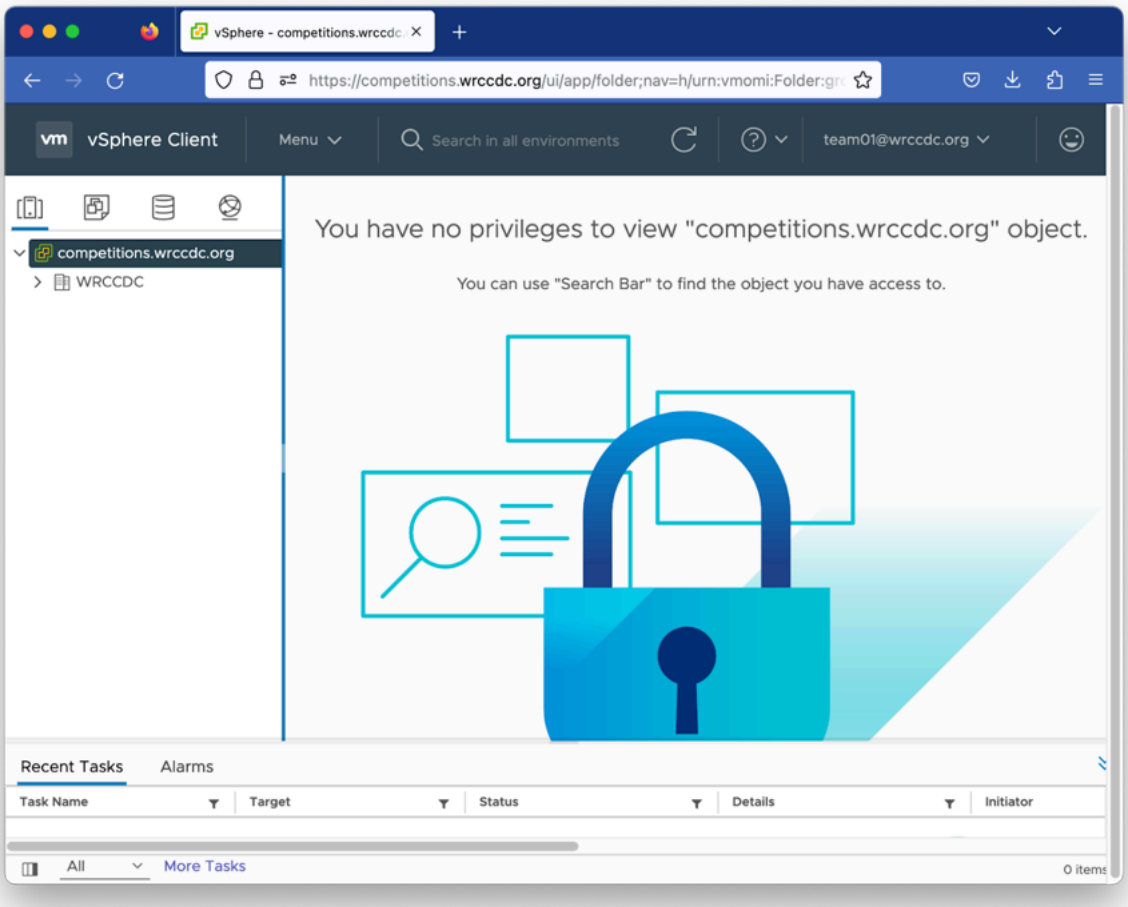
Limited vCenter Access

We are providing limited access to vCenter to teams. You can connect to the vCenter at https://competitions.wrccdc.org. This includes potentially taking snapshots, restoring snapshots, power cycling the system, and general management of the system. **Current Console Access is not enabled and vCenter access for Qualifiers MAY NOT be available. Do not plan as a strategy for console access!**

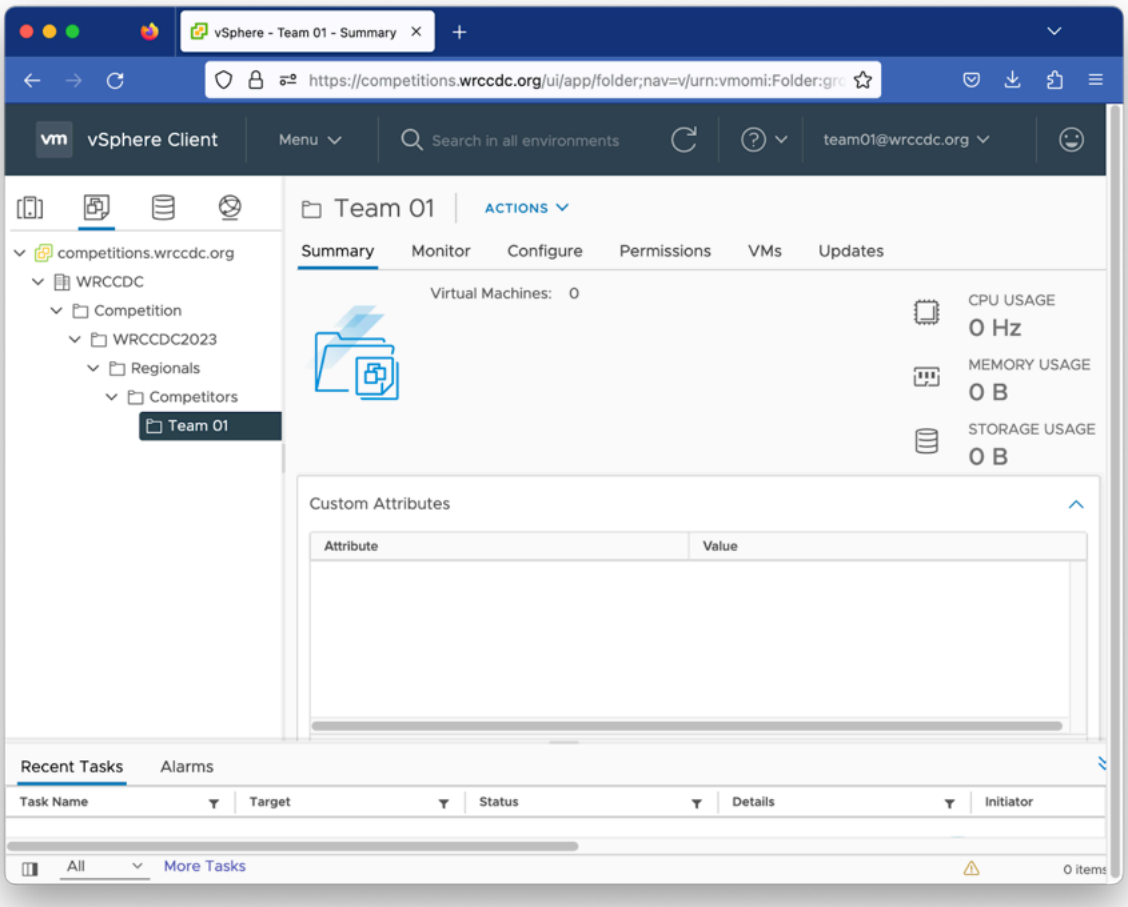
Your competition username: teamXX (where team XX is 01…04..10, 25, etc)

Your password: (See Password Document)

Upon initial access you may not be in the right view, Please click the “VM Icon” or Go to Menu and click “VMs and Templates”



Once logged in you must click the “VMs and Templates Tab”, from there you can navigate to your proper folder and see your VMs:



Support and Tickets

Discord

Discord may be used during the competition as a means of communicating to Black Team, Orange Team, and White Team. It is a means of communicating between your team securely and an easy way to share files to your team and competition organizers. You will not be required to leave after the competition, and you may use this Discord server to freely chat between schools and teams and participate in other events we have. Your team will be unassigned from you after the competition.

As part of this packet (or sent separately at the same time) you will have been given thirteen tokens (Up to 12 Team Members, including alternates and 1 for the team's coach. These may be used to register you into your team role in the competition discord.

Steps to Join and Setup Discord:

1. Join Discord using a personal account or one generated for the competition provided by the organizers.
2. Read the instructions in #welcome.
3. Set your role using !usekey in #role-selector
4. Begin using your team channel
5. You are joined!

Ticket Service

Our support system is available at https://tickets.wrccdc.org

If you have any issues during competition, or want to request consultation services, please do it via this portal. Once in, select your team name from the top right corner and create tickets from this group. There are tags for common issues including password changes, hardware issues, and verification of scores. Black team will follow up with these issues as soon as they are able.

Your competition username: teamXX (where team XX is 01…04..10, 25, etc)

Your password: (See Password Document)

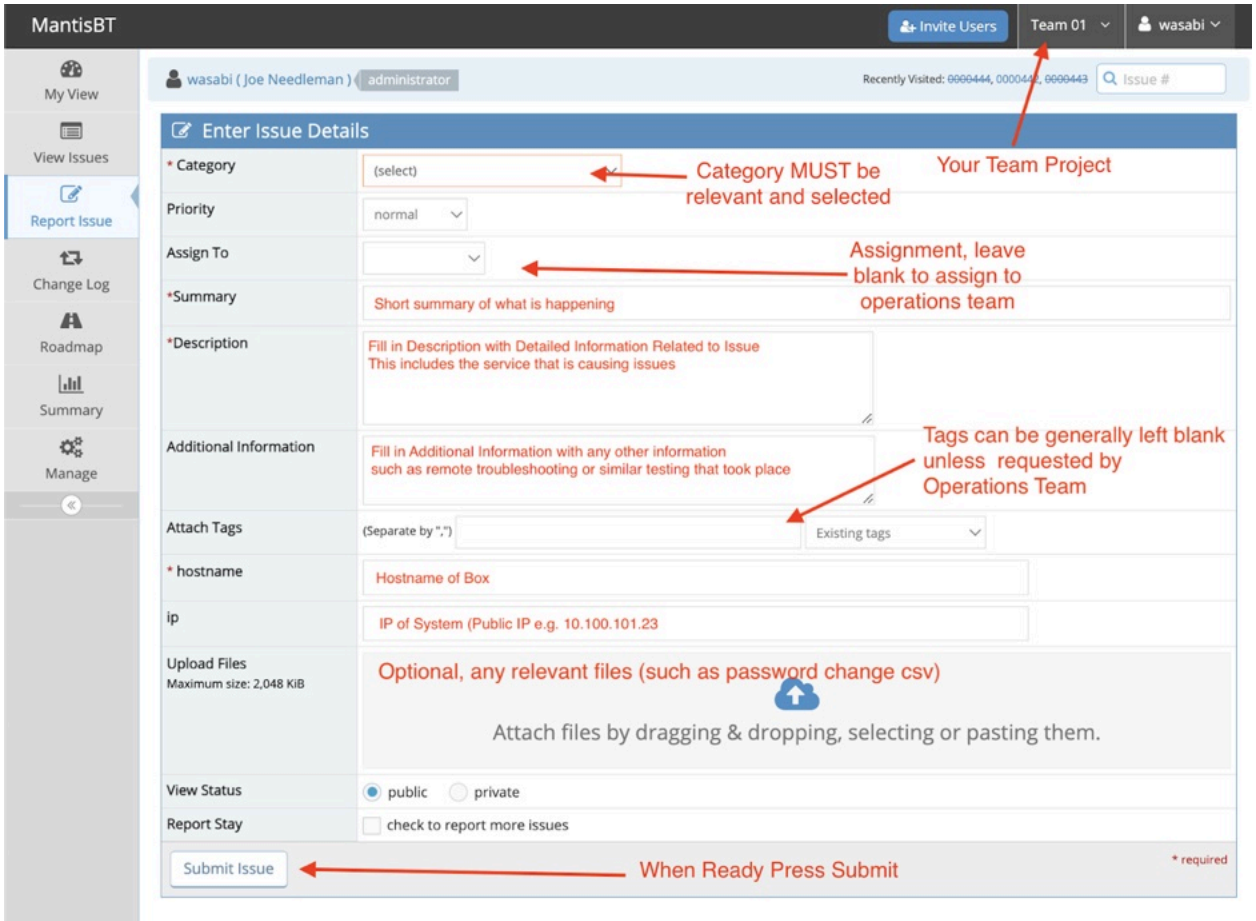
Submitting Tickets (Blue Team)

We utilize Mantis Ticketing System, to authenticate you need to use the same credentials you use for logging into Global Protect and vCenter. You can be on VPN and off VPN to access tickets.

The ticket system is located at https://tickets.wrccdc.org. Please use your existing credentials to log in.

Once logged in you can submit and monitor tickets during the competition. *In the final 5 minutes of competition, new tickets will NOT be addressed.* Please ensure everything is submitted in a timely manner.

Make sure all tickets have the proper public IP address (e.g. 10.100.1XX.YY) and proper hostname. You can get the hostname from checking on the host or via vCenter. Invalid entries will not be accepted by the ticketing system.



Common Service Requests

- Service Scoring Validation
  - 0 points, but we'll cut you off if you abuse it
  - If you believe your service is working 100% correctly and you want us to verify the check, file this ticket. If it's used frequently without additional consultations, we will require a Service Scoring Check ticket at minimum.
- Service Reset / Scrub
  - 60 points
  - We will reset your box to start of competition state and notify you when it is ready
- Scoring Service Check
  - 10 points
  - Have black team provide additional context surrounding the service check (details on the failure)
- Black Team Phone Consultation
  - 100 points
  - Have black team diagnose your issue over the phone with you
- Black Team Hands on Consultation

    - 200 points
    - Have black team gain access to your box to investigate and describe the issue to you, attempting to to fix things along the way
- Orange Team Verification/Questions
    - 10 points
    - Have orange team respond to you about how a score or service was performed
- Proxy
    - No point cost
    - Done as best effort to add to allowlists.

Inject Scoring

Inject scoring takes place in the Scoring Engine. You will need to track submissions to ensure they're complete and on time!

All injects are timed. They will show their completion time in ISE. There will also be a "Reject" Time. The "Reject" time will be the time at which submissions will no longer be accepted. We try to make the reject and completion time the same. Late injects will not be accepted.

All Judging is final. It will take us about a day or two to calculate scores and provide them to the competition organizers at which time finalists will be published. It is our intent to share the scoring rubrics to their teams. Teams will not receive other teams rubrics.

This will be the only notice for inject scoring guidelines. Points will be deducted for not following these rules.

File Names:

File names must be in the following format:

- Inject number must be first
- Team Numbers Only - **DO NOT** mention your School
- Underscores as spacing
- If the inject is a single digit, pad with a leading zero
- All lowercase letters

File types must be PDF only. No other file formats will be accepted.

Example of file naming convention:

- inject04_team13.pdf

Citing Sources:

When revising an existing work, such as editing a template found online, you **must** cite the source. The format of the source is not important and does not need to be standardized (MLA, APA, EIEIO, etc.). A reference URL is good enough.

Example reference, or “reference”, if you will:

> Our team was able to find a sample policy from the following site: https://templates.office.com/

If you follow these submission guidelines, you will do just fine. Good luck team!

Use of Artificial Intelligence sites such as ChatGPT or Google Bard will cause you to lose ALL points for that inject. We will be checking injects against several online portals setup by OpenAI, Alphabet, and other Universities against your submissions. IF they post a result of 20% or greater being fake, the inject will be thrown out.

Manual Scoring (Orange Team)

There will be services that are scored manually. These can be blogs, file services, calculators, spreadsheets, Outlook Web Access (OWA) portals, Games, etc. Orange Team members will be checking these. What is legitimate to check? Typically, we try to stick to the main applications that you would see in a real business. For example, if you notice a FTP site that requires authentication, that could be scored whereas one that allows anonymous access would not. Same with remote access. A SSH connection could be scorable whereas a Telnet Access Session would not. We try to make this as “common sense” as possible. Typically, the Orange team will contact you via Tickets, or Discord Chat (if available) to let you know that something is down. Orange Team points are scored based on number of checks attempted and do not account for more than 10% of overall scored points.

Orange Team Services (Initial)

Discord Voice - Customers, Staff, and Vendors will use this resource to communicate with you for requests. These will be scored.

AD Accounts - We have several of our staff that may be requesting to have remote access due to many circumstances. They are working on various projects and will need access. Please help them. They will be in touch with you via your in-scope phone.

Other Non-AD Users may inquire about their passwords not being set properly. Please assist them with this.

There are several non-scored services that will be used. If a service goes down, a team (Orange) member will be in touch with you to notify you of it being offline. They will work with you to provide you as much information as possible to correct it.

Orange team will also request “business operations”, and facilitate testing team’s handling of this, and can engage with the team over text, vc, or any other suitable method of

communication. These business operations can include, but is not limited to, requesting teams to complete administrative tasks, ensuring the integrity of services, and emulating a customer.

We are developing applications on several platforms. If you block them off, you may be notified that they are down. If they are, you will need to restore them as quickly as possible.

We are experimenting with an ordering Chat Bot. Please ensure that it is functional.

Other circumstances may arise. They will be communicated to you via your In-Scope Phone

NOTE - RED TEAM does not have access to your Discord Server. They will ***not*** impersonate the Orange Team during this competition. They will not be able to use the DISCORD Printer either.

If you interact with the Red Team, it will be on your VMs that you are managing ONLY!

Point Deductions (Red Team)

If in case your environment or one of your applications / systems is compromised, points will be deducted as outlined below. This is direct from National Scoring Guidelines.

Successful Red Team actions will result in penalties that reduce the affected team's score. Red Team actions include the following (penalties may be different than listed below):

- Obtaining root/administrator level access to a team system: **-100 points**
- Obtaining user level access to a team system (shell access or equivalent): **-25 points**
    - If standard users can be escalated to Root/Administrator Privilege, this is an additional **-100 point deduction.**
- Recovery of user IDs and passwords from a team system (encrypted or unencrypted): **-50 points**
    - For example, a user list, Active Directory with Hashes, SAM file, Shadow File
- Recovery of one or more sensitive files or pieces of information from a team system (configuration files, corporate data, etc.): **-25 points**
- Recovery of customer credit card numbers: **-50 points**
- Recovery of personally identifiable customer information (name, address, and credit card number): **-200 points**
- Recovery of encrypted customer data or an encrypted database: **-25 points**
    - -**25 points** additional if database can be unencrypted

Red Team actions are cumulative. For example, a successful attack that yields a system breach that causes a dump of active directory hashes followed by the decryption of said hashes leading to a user login finalized by a privilege escalation to Administrator that provides access to an encrypted database with customer data that in turn allows for the compromise of privilege information of customers' addresses and telephone numbers would be a net deduction of:

- -100 for System Breach
- -50 for AD hash dump

- -25 for Database Recovery
- -25 for Database Decryption
- -25 for Customer Data Breach
- -200 Points for PII loss

Total Deduction from one Incident would be: -425 Points.

Red Team actions are scored on a per system and per method basis – a buffer overflow attack that allows the Red Team to penetrate a team's system will only be scored once for that system; however, a different attack (Vulnerability) that allows the Red Team to penetrate the same system will also be scored. Only the highest level of account access will be scored per attack – for example, if the Red Team comprises a single user account and obtains root access in the same attack the penalty will be -100 points for root level access and not -125 points for root and user level access. Please note the point values described above are examples – actual penalty points may be adjusted to match the competition environment.

Red Teams can also execute additional malicious action based on their access. Attacks such as defacing websites, disabling or stopping services, adding/removing users, and removing or modifying files are permitted and may occur. This can affect service scoring and is legal Red Team Activity. They can reboot servers. Red Team needs to provide proof of breach or data compromise along with a date / time stamp for point deduction.

Incident Reports (Red Team)

You can provide submission to our incident response form to get a percentage of your lost points back if you accurately identify the attack.

Remote Site Judges and Connections

YOU MUST HAVE A REMOTE SITE JUDGE. If you do not you will be asked to stop playing. In addition you must have a maximum of 8 connections to the VPN at any given time. If we see more we will reach out to your remote site judge. Repeatedly going over 8 connections WILL RESULT in a ban from the event.

Proxy Server

The proxy server is in place to ensure similarity to nationals and limit access to resources not authorized (or previously approved). You are not required to use it however your in-pod systems will be restricted from communicating outbound. We block HTTP, HTTPS, FTP, and SSH access out of the pods. To use HTTP, HTTPS, or FTP you must enable the proxy.

To use the proxy head to http://proxy.wrccdc.org or 10.0.0.211. There is an information page served to allow you to download the PEM and DER certificates. These must be installed into the respective root certificate authority of your system or browser (for example Firefox uses its own certificate manager). This server also can act as a caching mirror apt, apk, pacman, rpm, dnf, and other package managers. This can speed up access time for files during the competition significantly.